ASSIGNMENT-2 : TYPES OF HACKERS

GUIDED BY : VIPUL MISTRI

Q.1 Create a table listing and describing the main types of hackers (such as white hat, black hat, grey hat, script kiddie, hacktivist, and state-sponsored) and give one real-world example or news headline for each type.

ANS :

Hacker Type	**Primary Motivation**	**Core Characteristics & Skill Level**	**Real-World Example / Headline**
White Hat *(Ethical Hacker)*	Security enhancement, vulnerability disclosure, and system defense.	Highly skilled cyber security professionals who act with explicit permission from system owners. They conduct penetration testing, bug bounty hunting, and security audits to patch security flaws before malicious entities find them.	**"White-Hat Hacker Awarded $100,000 for Finding Zero-Day Vulnerability in Apple iOS"** *(Security researchers routinely identify and responsibly report critical system vulnerabilities via official Bug Bounty programs).*
Black Hat *(Malicious Hacker)*	Personal financial gain, data theft, extortion, or intentional disruption.	Skilled threat actors operating illegally without authorization. They exploit system vulnerabilities to deploy ransomware, steal sensitive personal or financial records, and sell stolen intellectual	**"Colonial Pipeline Pays $4.4 Million Ransom Following Devastating Cyberattack"**

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		property on dark web marketplaces.	*(Black-hat criminal operators targeted critical energy infrastructure using ransomware, forcing a major U.S. pipeline shutdown).*
Grey Hat	Curiosity, notoriety, or opportunistic discovery (morally ambiguous).	Skilled individuals who locate security vulnerabilities without authorization, but without malicious intent to steal or destroy data. They may notify the target company after breaching the system or ask for a fee/reward to reveal how they did it.	**"Researcher Infiltrates Enterprise Server Without Authorization to Disclose Open Cloud Database"** *(A grey hat bypasses security controls without permission, then emails the site administrator warning them of the flaw).*
Script Kiddie	Seeking thrill, peer recognition, or	Low-skilled individuals who rely on pre-written scripts, automated	**"Teenager Arrested After Renting Cyber**

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	self-amusement.	hacking toolkits, or rented botnets created by skilled hackers. They lack an in-depth understanding of underlying network protocols or code structures.	**Booter Service to Take Down School District Website"** *(An unsophisticated actor uses a paid Distributed Denial of Service (DDoS) tool to crash local educational portals).*
Hacktivist	Social, political, ideological, or environmental causes.	Highly motivated groups or individuals using digital disruptions (such as web defacements, DDoS attacks, or hack-and-leak operations) as a form of digital protest or civil disobedience to promote a specific ideology.	**"Pro-Palestinian Group Handala Hack Targets Israeli Infrastructure in Hack-and-Leak Campaign"** *(Hacktivists execute data leaks and website defacements to bring global attention to active political or*

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			geopolitical conflicts).
State-Sponsored *(APT Actor)*	Geopolitical espionage, state-level economic theft, and cyber warfare.	Nation-state-backed Advanced Persistent Threat (APT) groups with massive resources, advanced custom exploits, and long-term targets. They conduct covert operations aimed at government agencies, defense contractors, and critical infrastructure.	**"FBI Links $6.3 Billion Crypto Heist to North Korean Hacking Group TraderTraitor"** *(State-backed threat groups leverage high-level cyber operations to bypass advanced security controls and fund nation-state initiatives).*

Q.2 Pick any popular app you use (Instagram, Paytm, Zomato, etc.) and describe a scenario for each type of hacker: how would a white hat, black hat, and grey hat hacker interact with this app? Write 2-3 sentences per scenario.

ANS : Here is how a White Hat, Black Hat, and Grey Hat hacker would interact with **Instagram**:

- **White Hat Scenario:** An ethical security researcher discovers a broken object level authorization (BOLA) flaw in Instagram's direct messaging API that could allow unauthorized access to private chats. Working through Meta's official Bug Bounty program, they document the technical reproduction steps, report it

confidentially to Meta’s security team, and receive a bounty while allowing Instagram engineers to patch the vulnerability before public disclosure.

- **Black Hat Scenario:** A cybercriminal exploits an unpatched session-hijacking vulnerability on Instagram's web application to execute an automated credential harvesting campaign. They bypass two-factor authentication, silently seize control of high-profile creator accounts, and publish fraudulent cryptocurrency phishing links in stories to defraud millions of followers for immediate financial gain.

- **Grey Hat Scenario:** An independent hacker scans Instagram’s public endpoints without authorization and finds a configuration flaw that leaks user email addresses linked to private accounts. Rather than exploiting it for profit or reporting it through official channels, they post a redacted proof-of-concept video on social media or reach out directly to Instagram's CEO demanding a consulting fee to explain how to fix it.

Q.3 Given the following user actions, classify each as white hat, black hat, grey hat, or other: reporting a bug to the company, stealing user data, testing your own account for vulnerabilities, defacing a website for a political cause, and participating in a government cyber operation. Hint: Use your table from Task 1 for quick reference.

ANS : Here is the classification for each user action based on hacker categories and intent:

- **Reporting a bug to the company: White Hat** — The user responsibly discloses security vulnerabilities directly to the platform owner so the flaw can be remediated safely.

- **Stealing user data: Black Hat** — Exfiltrating sensitive personal or financial information without authorization is a malicious and illegal act intended for personal, financial, or criminal gain.

- **Testing your own account for vulnerabilities: White Hat** (or **Other / Defensive Security**) — Assessing an account or system you explicitly own is a authorized self-audit to improve personal posture and security hygiene.

- **Defacing a website for a political cause: Hacktivist** (a specialized subtype of **Black Hat / Grey Hat**) — Bypassing access controls to alter web content for political or ideological expression without the site owner's authorization.

- **Participating in a government cyber operation: State-Sponsored** (or **APT Actor / Defence Operative**) — Conducting intelligence, defense, or counter-offensive cyber actions under the explicit authorization and command of a nation-state entity.

Q.4 Interview-style: Imagine you are explaining to your non-technical friend the difference between a white hat and a black hat hacker using a cricket match analogy. Write your explanation in 4-5 sentences.

ANS : Think of a cricket match where the system is the pitch and stadium. A **white hat hacker** is like an official pitch curator or security marshal—they test the pitch, spot dangerous cracks, and report them to the groundskeeper so nobody gets hurt during the game. A **black hat hacker**, on the other hand, is like a corrupt pitch-tamperer who sneaks onto the field at night to dig hidden holes or fix the match for their own financial gain. Both know every technical detail of how the pitch works, but while the white hat uses that knowledge to protect the players and keep the game fair, the black hat uses it to wreck the match and rob the spectators.